

SE1 Summative 2 Mock Exam: Modules 3 and 4

50 items: choice, dropdown matching, and typed lists. Answer key at the end.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Part 1: Multiple Choice and True or False

1. Which Scrum role is the facilitator of the scrum development process?

- A. Stakeholder
- B. Scrum Master
- C. Development Team
- D. Product Owner

2. Data modeling is used for what?

- A. Scheduling sprints
- B. Defining business requirements for a database
- C. Designing hardware
- D. Writing test cases

3. Which fact finding technique collects data from large groups?

- A. Questionnaires
- B. Interviews
- C. Research
- D. Observation

4. What does modeling a system mean?

- A. Writing the final code
- B. Identifying its characteristics, states, and behavior using a notation
- C. Buying hardware
- D. Deleting old files

5. In which model is testing done only after the development phase?
- A. Waterfall
 - B. XP
 - C. Agile
 - D. Scrum
6. How long is sprint planning for a 2-week sprint?
- A. Two hours
 - B. Fifteen minutes
 - C. Four hours
 - D. One and a half hours
7. Agile emphasizes individuals and interactions over processes and tools.
- A. True
 - B. False
8. According to the Agile principles, how often should working software be delivered?
- A. Once a year
 - B. Frequently, in weeks rather than months
 - C. Every few years
 - D. Only at the end of the project
9. Which DFD guideline says to check that every symbol is logically connected?
- A. Label the top
 - B. Use a template
 - C. Use only one symbol
 - D. Desk check all data flow diagrams
10. Which of the following is an example of a supporting process?
- A. Strategic management
 - B. Marketing and sales
 - C. Recruitment and technical support
 - D. Product manufacturing

11. In a DFD, data stores are represented by what?
- A. A circle
 - B. A rounded rectangle
 - C. An open rectangle
 - D. An arrow
12. In a DFD, which symbol denotes input and output elements such as a person or place?
- A. Rectangle (entity)
 - B. Open rectangle
 - C. Arrow
 - D. Rounded rectangle
13. In functional modeling, which element is outside the modeled system but interacts with it?
- A. Flow
 - B. Process
 - C. External entity
 - D. Store
14. Agile encourages teams to respond to changing requirements:
- A. Even late in development
 - B. Only before development starts
 - C. Never after requirements are approved
 - D. Only during planning
15. What is the primary purpose of Business Process Modeling (BPM)?
- A. To identify computer specifications
 - B. To determine employee salaries
 - C. To represent, analyze, and improve organizational processes
 - D. To install software applications
16. Extreme Programming (XP) is very helpful when what is true?
- A. There is no customer
 - B. The project has a fixed contract
 - C. Requirements never change
 - D. Customer demands or requirements constantly change

17. What does systems modeling use to conceptualize and construct systems?

- A. Programming languages only
- B. Financial statements
- C. Marketing plans
- D. Models and diagrams

18. What does each sprint start with?

- A. A retrospective
- B. A release
- C. A sprint review
- D. A sprint planning event that establishes a sprint goal

19. Which Scrum role represents the stakeholders and administers the scrum backlog?

- A. Tester
- B. Development Team
- C. Scrum Master
- D. Product Owner

20. How long is the sprint review for a two-week sprint?

- A. Fifteen minutes
- B. One and a half hours
- C. Two hours
- D. Four hours

21. Agile is built on the collaborative effort of which kind of teams?

- A. Self-organizing and cross-functional teams and their customers or end users
- B. Only managers
- C. Separate departments that never meet
- D. Only external contractors

22. Lean software development is adapted from which system?

- A. The Toyota Production System
- B. The Waterfall model
- C. The Agile Manifesto
- D. The Scrum guide

23. What does a functional block diagram describe?

- A. Hardware prices
- B. The functions and interrelationships of a system
- C. Only the database tables
- D. Employee records

24. In which XP phase do pilot launch, training, and production launch happen?

- A. Wrapping
- B. Analysis
- C. Planning
- D. Closure

25. What is MoSCoW used for?

- A. Writing code
- B. Prioritizing requirements with stakeholders
- C. Scheduling meetings
- D. Testing

26. In which XP phase does coding and unit testing happen?

- A. Wrapping
- B. Closure
- C. Execution
- D. Planning

27. Which Lean principle says the sooner the product is delivered without major defects, the sooner feedback comes?

- A. Defer Commitment
- B. Early Delivery
- C. Eliminating Waste
- D. Building Integrity

28. Which sequence correctly represents the three phases of Crystal Methodology?

- A. Chartering, Cyclic Delivery, Wrap Up
- B. Planning, Coding, Testing
- C. Analysis, Design, Deployment
- D. Feasibility, Development, Maintenance

29. What are system requirements?

- A. Only the price of software
- B. A legal document
- C. A list of users
- D. The required specifications a device must have to use certain hardware or software

30. Which type of business process creates the organization's primary value stream?

- A. Operational process
- B. Documentation process
- C. Management process
- D. Supporting process

31. What is a model in systems development?

- A. A simplified description of a system
- B. A complete copy of an existing system
- C. A collection of programming codes
- D. A list of hardware components

32. Which modeling approach represents the functions, activities, and operations within a system?

- A. Functional modeling
- B. Financial modeling
- C. Enterprise modeling
- D. Data modeling

33. Which pair of notations can be used for DFDs?

- A. ER and IDEF
- B. UML and BPMN
- C. Gane & Sarson or Yourdon & DeMarco
- D. Chen and Crow's Foot

34. Which system requirement is not flexible?

- A. RAM
- B. The operating system and disk space needed for installation
- C. CPU speed
- D. GPU

Part 2: Matching

35. Match the model with its description.

- A. Flows sequentially; customer sees the product at the end
- B. Only used for small hardware
- C. Incremental and iterative; customer sees the product early

Agile: _____

Waterfall: _____

36. Match the Scrum item with its time or size.

- A. 4 hours
- B. 15 minutes
- C. 1.5 hours
- D. 1 day
- E. 3 to 10 people
- F. 2 hours
- G. 30 minutes

Daily scrum: _____

Sprint planning (2-week sprint): _____

Sprint review (2-week sprint): _____

Sprint retrospective (2-week sprint): _____

Scrum team: _____

37. Match the business process type with its example.

- A. Corporate governance and strategic management
- B. Software testing only
- C. Purchasing, manufacturing, marketing, sales
- D. Accounting, recruitment, technical support

Management process: _____

Operational process: _____

Supporting process: _____

38. Match the fact finding technique with its description.

- A. Reports, bills, policy manuals, procedures
- B. Choose the right person and place to watch
- C. Published materials used to verify facts
- D. Discover misunderstandings and resistance; time consuming
- E. Guessing
- F. Data from large groups; open or closed

Interviews: _____

Record inspections: _____

Questionnaires: _____

Observation: _____

Research: _____

Part 3: Typed Lists

39. List the three questions asked in the sprint retrospective. (3 items)

1. _____
2. _____
3. _____

40. List the three Scrum roles. (3 items)

1. _____
2. _____
3. _____

41. List the four things that happen in each Crystal Cyclic Delivery cycle. (4 items)

1. _____
2. _____
3. _____
4. _____

42. List the four MoSCoW categories. (4 items)

1. _____
2. _____
3. _____
4. _____

43. List the seven principles of Lean software development. (7 items)

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

44. List the four Agile Development Values. (4 items)

1. _____
2. _____
3. _____
4. _____

45. List the two types of questionnaires. (2 items)

1. _____
2. _____

46. List the four elements of a functional model. (4 items)

1. _____
2. _____
3. _____
4. _____

47. List the two sets of system requirements. (2 items)

1. _____
2. _____

48. List the three types of business process. (3 items)

1. _____
2. _____
3. _____

49. List the four types of system modeling in the module. (4 items)

1. _____
2. _____
3. _____
4. _____

50. List any four design diagrams named in the module. (4 items)

1. _____
2. _____
3. _____
4. _____

Answer Key

No.	Answer	Why
1	B. Scrum Master	Ch. 1: Scrum roles
2	B. Defining business requirements for a database	Ch. 1: Data modeling
3	A. Questionnaires	Ch. 2: Fact finding
4	B. Identifying its characteristics, states, and behavior using a notation	Ch. 1: System models
5	A. Waterfall	Ch. 1: Agile vs Waterfall
6	C. Four hours	Ch. 1: Sprint planning
7	A. True	Ch. 1: Agile values
8	B. Frequently, in weeks rather than months	Ch. 1: Agile principles
9	D. Desk check all data flow diagrams	Ch. 1: DFD guidelines
10	C. Recruitment and technical support	Ch. 1: Business process types
11	C. An open rectangle	Ch. 1: DFD symbols
12	A. Rectangle (entity)	Ch. 1: DFD symbols
13	C. External entity	Ch. 1: Functional modeling
14	A. Even late in development	Ch. 1: Agile principles
15	C. To represent, analyze, and improve organizational processes	Ch. 1: BPM
16	D. Customer demands or requirements constantly change	Ch. 2: XP
17	D. Models and diagrams	Ch. 1: System models
18	D. A sprint planning event that establishes a sprint goal	Ch. 1: Scrum events
19	D. Product Owner	Ch. 1: Scrum roles
20	C. Two hours	Ch. 1: Sprint review
21	A. Self-organizing and cross-functional teams and their customers or end users	Ch. 1: What is Agile
22	A. The Toyota Production System	Ch. 2: Lean
23	B. The functions and interrelationships of a system	Ch. 1: Functional block diagram
24	D. Closure	Ch. 2: XP phases
25	B. Prioritizing requirements with stakeholders	Ch. 2: MoSCoW
26	C. Execution	Ch. 2: XP phases
27	B. Early Delivery	Ch. 2: Lean principles
28	A. Chartering, Cyclic Delivery, Wrap Up	Ch. 2: Crystal phases
29	D. The required specifications a device must have to use certain	Ch. 2: System requirements

	hardware or software	
30	A. Operational process	Ch. 1: Business process types
31	A. A simplified description of a system	Ch. 1: System models
32	A. Functional modeling	Ch. 1: Functional modeling
33	C. Gane & Sarson or Yourdon & DeMarco	Ch. 1: DFD guidelines
34	B. The operating system and disk space needed for installation	Ch. 2: System requirements
35	Agile = C; Waterfall = A	Ch. 1: Agile vs Waterfall
36	Daily scrum = B; Sprint planning (2-week sprint) = A; Sprint review (2-week sprint) = F; Sprint retrospective (2-week sprint) = C; Scrum team = E	Ch. 1: Scrum events
37	Management process = A; Operational process = C; Supporting process = D	Ch. 1: Business process types
38	Interviews = D; Record inspections = A; Questionnaires = F; Observation = B; Research = C	Ch. 2: Fact finding
39	What went well during the sprint?, What did not go well?, What could be improved for better productivity in the next sprint? (any order)	Ch. 1: Sprint retrospective
40	Scrum Master, Product Owner, Scrum Development Team (any order)	Ch. 1: Scrum roles
41	Team updates and refines the release plan; Implements a subset of the requirements through one or more program test integrate iterations; Integrated product is delivered to real users; Review of the project plan and adopted development methodology (in order)	Ch. 2: Crystal cyclic delivery
42	Must have; Should have; Could have; Won't have (in order)	Ch. 2: MoSCoW
43	Eliminating Waste; Amplifying Learning; Defer Commitment; Early Delivery; Empowering the Team; Building Integrity; Optimize the Whole (in order)	Ch. 2: Lean principles
44	Individuals and interactions over processes and tools, Working	Ch. 1: Agile values

	software over comprehensive documentation, Customer collaboration over contract negotiation, Responding to change over following a plan (any order)	
45	Open-ended, Closed-ended (any order)	Ch. 2: Questionnaires
46	Process, Store, Flow, External Entity (any order)	Ch. 1: Functional modeling
47	Minimum, Recommended (any order)	Ch. 2: System requirements
48	Management Process; Operational Process; Supporting Process (in order)	Ch. 1: Business process types
49	Functional Modeling; System Architecture; Business Process Modeling; Enterprise Modeling (in order)	Ch. 1: System models
50	Data flow diagrams, Structured charts, Decision trees, Block diagrams (cycle of operations) (any order)	Ch. 1: Design diagrams